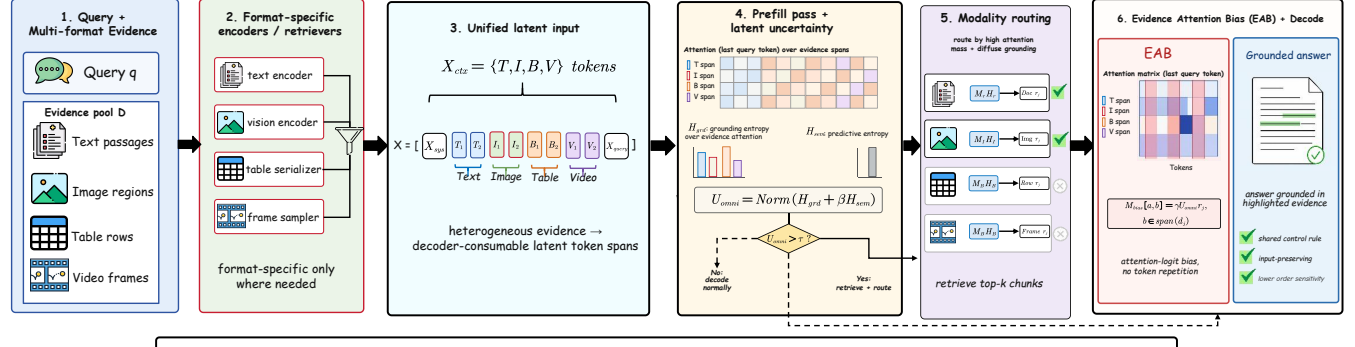

1. Query + Multi-format Evidence
Query q
Evidence pool D
Text passages
Image regions
Table rows
Video frames
2. Format-specific encoders / retrievers
text encoder
vision encoder
table serializer
frame sampler
format-specific only where needed
3. Unified latent input
$X_{ctx} = \{T, I, B, V\}$ tokens
X = [ $X_{sys}$ $T_1$ $T_2$ $I_1$ $I_2$ $B_1$ $B_2$ $V_1$ $V_2$ $X_{query}$ ]
Text
Image
Table
Video
heterogeneous evidence → decoder-consumable latent token spans
4. Prefill pass + latent uncertainty
Attention (last query token) over evidence spans
T span
I span
B span
V span
$H_{grd}$ grounding entropy over evidence attention
$H_{sem}$ predictive entropy
$U_{omni} = Norm(H_{grd} + \beta H_{sem})$
$U_{omni} > \tau$ ?
No: decode normally
Yes: retrieve + route
5. Modality routing
route by high attention mass + diffuse grounding
$M_1 H_1$
$M_2 H_2$
$M_3 H_3$
$M_4 H_4$
retrieve top-k chunks
6. Evidence Attention Bias (EAB) + Decode
EAB
Attention matrix (last query token)
T span
I span
B span
V span
Tokens
$M_{eab}[a,b] = \gamma U_{omni} r_j$, $b \in span(d_j)$
attention-logit bias, no token repetition
Grounded answer
answer grounded in highlighted evidence
shared control rule
input-preserving
lower order sensitivity
$T_1$ $T_2$ $I_1$ $I_2$ $B_1$ $B_2$ $V_1$ $V_2$ Latent evidence-token spans
retriever /index
attention-logit bias
conditional decision
data flow